

BIKE&SAFE DATASET

Title: Bike&Safe

Abstract: A model to support cyclist accident prevention in smart cities. This dataset collects data from the accelerometer, gyroscope and magnetometer sensors of the smartphone positioned on the handlebar of the bicycle.

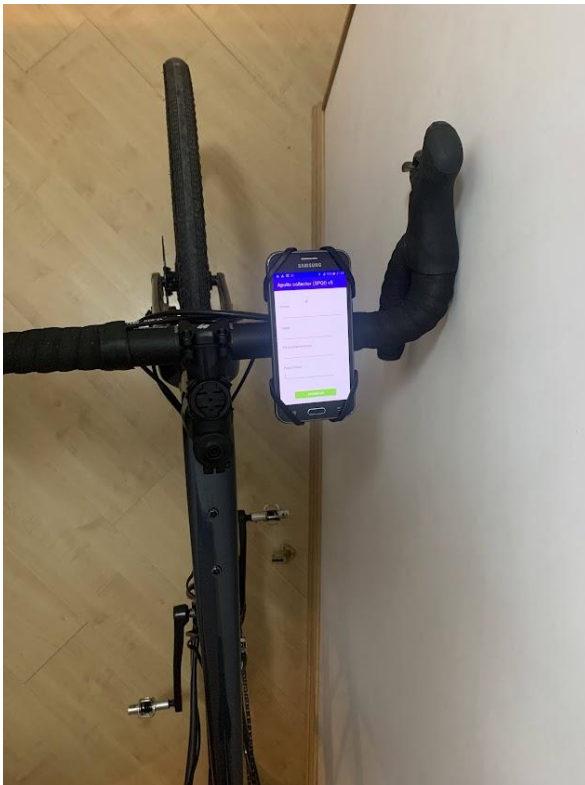


Figure 1: App image – Running

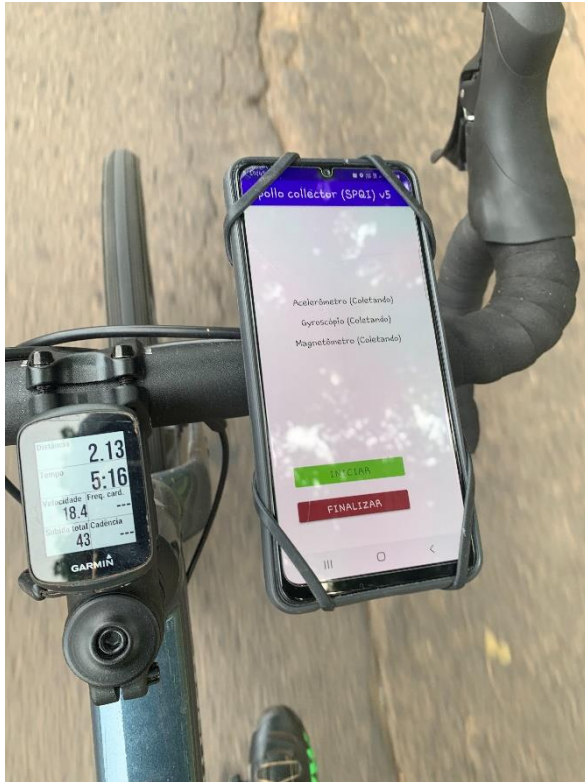


Figure 2: Photo on first route

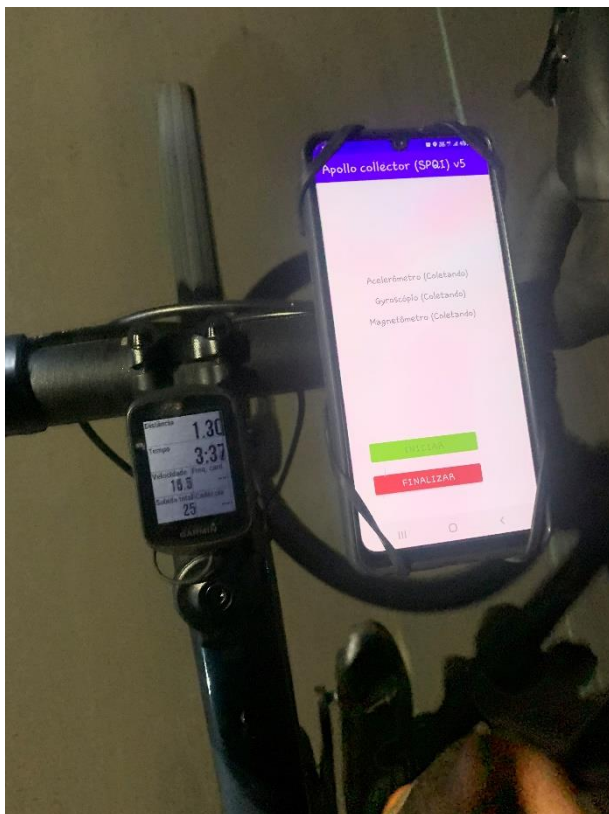


Figure 3: Photo on second route

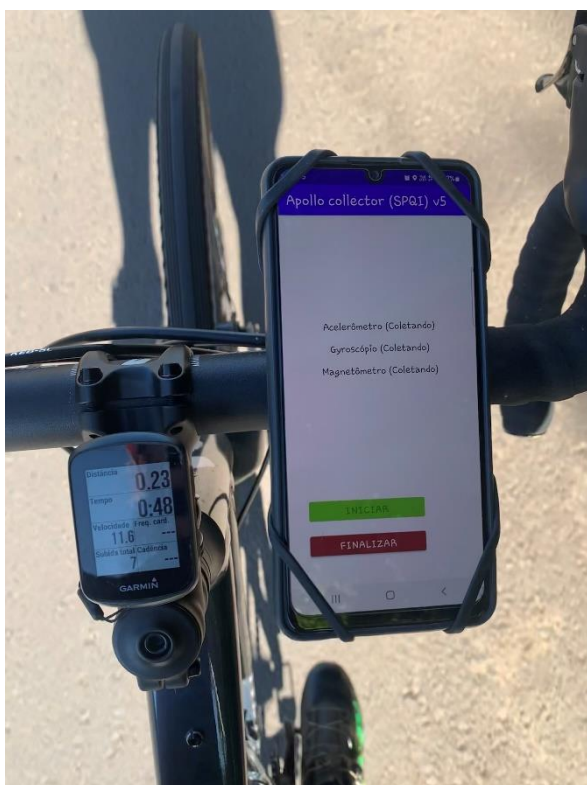


Figure 4: Photo on third route

First route:

Starting the ride on a known place on the city of Dois Irmãos in Brazil and getting to the city of Ivoti in Brazil, via BR-16 highway.

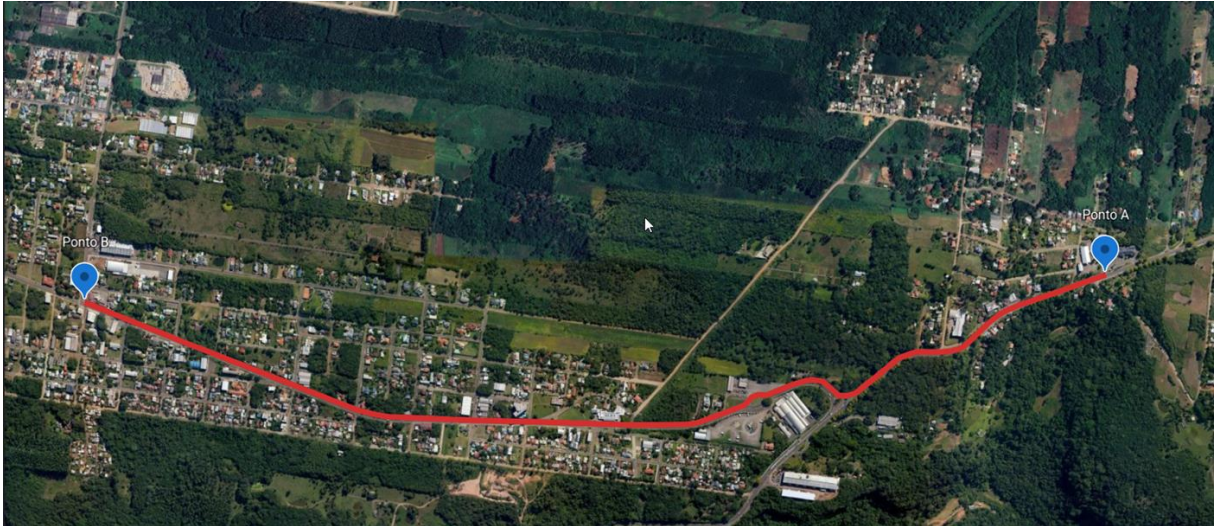


Figure 5: First route map

Route link: <https://bit.ly/3NsJ3vx>

Second route:

Starting the ride at the beginning of the downtown of Ivoti city in Brazil, and ending it at the end of the downtown

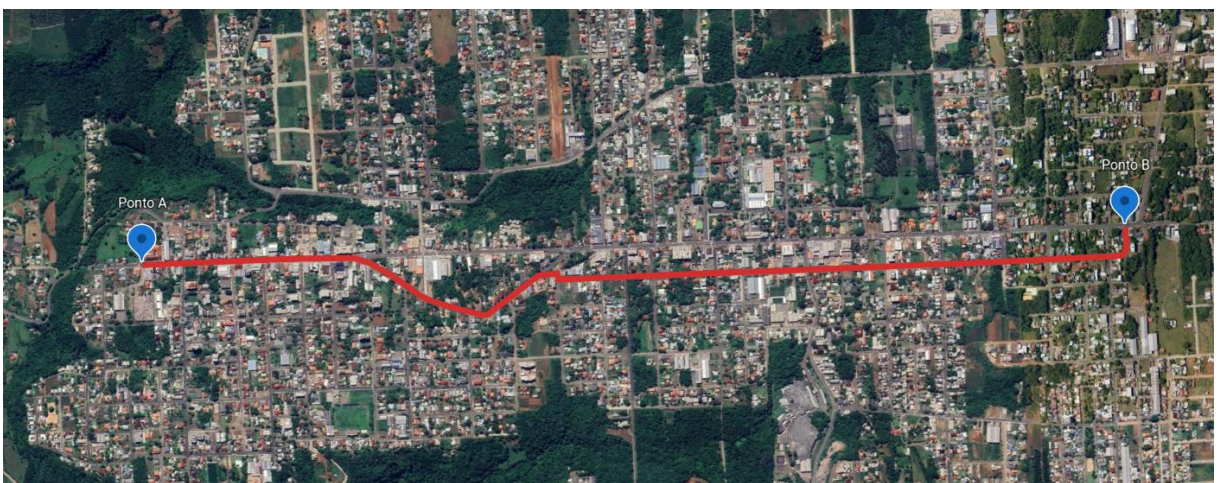


Figure 6: Second route map

Route link: <https://bit.ly/3DR3jUk>

Third route:



Figure 7: Third route map

Route link: <https://bit.ly/3TXFIMU>

Number of samples: 12

Number of Instance: 811781

Data Set Characteristics: Sequential Data, Time Series Data, Multivariate Data, Correlation

Area: Traffic Safety

Attribute Characteristics: Integer, Double

Number of Attributes: 5

Relevant Information:

1. The dataset collects data from an Android smartphone which is positioned in the center of the walker.
2. The data sensors collected are accelerometer, gyroscope and magnetometer.
3. The sampling frequency of the sensors is DELAY_FASTEST - 20 milliseconds.
4. Data format: CSV

Dataset information:

1. The dataset was recorded in three different scenarios. Each scenario goes from point A to point B, via roads.
2. Each scenario was recorded 3 times.
3. Each scenario was recorded 20 times.
4. Each recording generated four files:
 - i. Accelerometer.
 - ii. Gyroscope.
 - iii. Magnetometer.
 - iv. GPS.
5. Three first files contain the following information
 - i. Timestamp, Id, x axis, y axis, z axis.
6. GPS file contain the following information
 - i. Timestamp, ID, Latitude, Longitude, Speed.

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